SHWETA VIJAY KULKARNI

Full-stack AI Engineer

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Professional Summary

Full-Stack AI & React.js Developer with 3 years of hands-on experience bridging the gap between scalable web architectures (React, FastAPI, Node.js) and production-grade Generative AI systems, including multi-modal RAG pipelines and secure data infrastructures.

Technical Skills

AI & LLM Orchestration

Retrieval-Augmented Generation (RAG), GPT-4o (Azure AI Inference SDK / GitHub Models), SentenceTransformers (all-MiniLM-L6-v2), FAISS, Source Attribution, Prompt Engineering

Backend & Database

Python (FastAPI, Uvicorn), Node.js,MongoDB (motor async driver), Redis (Caching), PostgreSQL, RESTful APIs, JWT Auth (python-jose, bcrypt)

Frontend Development

React.js, TypeScript, React Router, Redux, Context API, JavaScript (ES6+), Tailwind CSS, Material UI (MUI), Bootstrap 5

DevOps & Data Pipeline

Docker, Docker Compose, Nginx, PyPDF, OCR Document Parsing, ETL Workflows, Git, Postman

FEATURED AI PRODUCTION PROJECT

Enterprise Multi-Modal RAG Chatbot Platform

- **Full-Stack AI Architecture:** Built an asynchronous React.js and FastAPI platform, integrating **GPT-4o via Azure AI Inference SDK** for high-performance inference.
- **Advanced RAG Pipeline:** Deployed a multi-modal ingestion system (pypdf/OCR) with **SentenceTransformers** and **FAISS** for real-time vector embeddings and semantic search.
- **State & Performance Ops:** Engineered MongoDB (motor) for secure JWT chat histories and integrated **Redis caching** to slash query retrieval latency.
- **UI & Containerized DevOps:** Crafted a responsive React UI (React Markdown) and orchestrated the entire multi-service ecosystem using **Docker Compose**.

Professional Experience

React.js / Full-Stack AI Engineer, 02/2025 – 02/2026 | Pune

Techsolix Pvt Ltd

- Designed and developed production web applications using MongoDB, Express, React, and Node.js, engineering modular component libraries that cut engineering layout cycles by 25%.
- Re-architected client-facing user interfaces utilizing Tailwind CSS and Material UI, driving a verified surge in user retention and direct positive stakeholder feedback.
- Built and optimized end-to-end ETL workflows and asynchronous data pipelines using Python, streamlining cross-platform data synchronization for high-velocity datasets.
- Executed seamless database migrations and structural schema upgrades on live staging environments with near-zero system downtime.

Software Developer, Smart Ship-Hub 10/2024 – 01/2025 | Pune

- Engineered real-time backend data engines by executing highly optimized, non-blocking CRUD operations via Fastify, drastically improving data throughput for a core marine logistics dashboard.
- Diagnosed and resolved critical race conditions and bugs within active, live-streaming data feeds, maximizing software uptime and platform stability.
- Developed and deployed an asset-optimized web application utilizing React.js and TypeScript, slashing bundle sizes to dramatically improve core web vitals.

Project Intern, Skillhub IT solutions 10/2023 – 09/2024
Aurangabad

- Maintained production-level web applications built on React.js, implementing lazy loading and code-splitting practices to optimize interaction metrics.
- Collaborated closely with cross-functional product teams to translate high-fidelity UI/UX requirements from Figma into responsive, accessible HTML5/CSS3 layouts.
- Supported core database administration tasks, successfully managing automated backups, schema validation scripts, and system updates.

Education

Master of Business Administration (MBA),
Shreeyash Engineering and Management Institute